

UNIVERSIDADE SÃO FRANCISCO
Graduate Program in Health Data Science

Speech Analytics and Natural

Language Processing Applied to Clinical Communication: A Scoping Review

Principal Investigator:

- Vicente Idalberto Becerra Sablón Permanent Faculty, Graduate Program in Health Sciences, Universidade São Francisco

Research Team:

- Rafael Baena Neto — MSc student, Graduate Program in Health Data Science, Universidade São Francisco
- Fabrício Evangelista dos Santos — Undergraduate student, Computer Engineering, Universidade São Francisco
- Lucas Kenzo Terazzin Ida — Undergraduate student, Computer Engineering, Universidade São Francisco
- Murilo Nogueira Minutti — Undergraduate student, Computer Engineering, Universidade São Francisco

Dear Editor-in-Chief,

We are pleased to submit the manuscript entitled "Speech Analytics and Natural Language Processing Applied to Clinical Communication: A Scoping Review" for consideration for publication in the International Journal of Medical Informatics as a Review article.

SCOPE AND CONTRIBUTION

Clinical communication — particularly the anamnesis encounter — constitutes one of the richest yet least analytically exploited data sources in healthcare. Recent advances in Automatic Speech Recognition (ASR), Natural Language Processing (NLP) and vocal biomarker analysis have expanded the technical capacity to extract structured information from spoken clinical interactions.

Despite this progress, few studies operationalise the clinical interaction itself as a source of speech-derived measurable variables, particularly communicational and behavioural markers that can inform adherence monitoring and care quality assessment.

This scoping review systematically maps the evidence on Speech Analytics and NLP applied to clinical communication (2014–2026), following PRISMA-ScR guidelines (Tricco et al., 2018) and JBI methodology (Peters et al., 2020).

The corpus of 24 included studies — drawn from PubMed/MEDLINE and supplemented by backward snowballing — was organised into four thematic axes: (A) clinical NLP and biomedical entity extraction; (B) automated clinical documentation and ASR; (C) physician-patient communication; and (D) vocal biomarkers and multimodal analysis. The review confirms a consistent gap between the documented technical capabilities and the operationalisation of the anamnesis as a structured analytical unit, and proposes a research agenda for prospective studies integrating Speech Analytics with structured clinical communication analysis.

WHY IJMI

The International Journal of Medical Informatics is the primary venue for this work: the four thematic axes span NLP-based clinical information extraction, ASR-driven documentation systems, and computational approaches to clinical communication — all central to IJMI's scope. The confirmed gap identified (clinical interactions rarely operationalised as speech-derived variables) maps directly onto the journal's longstanding interest in the clinical translational challenges of health data science.

DECLARATIONS

This manuscript has not been published previously and is not under consideration for publication elsewhere. All authors have approved the final version and agreed to its submission.

This study is a scoping review based exclusively on published scientific literature; no primary data were collected and no ethical approval was required (CNS Resolution No. 510/2016).

Article processing charges (APC) are covered under the Open Access Read & Publish Agreement between CAPES (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Brazil) and Elsevier for the period 2026–2028. The corresponding author's institutional affiliation (Universidade São Francisco, Bragança Paulista, SP, Brazil) is eligible under this agreement.

Supplementary materials submitted with this manuscript:

- S1 — Complete electronic search strings by database (PubMed/MEDLINE)
- S2 — Standardised data extraction form (CSV)

We thank the editorial team for their time and consideration.

Sincerely,

Rafael Baena Neto
on behalf of all co-authors

